



wergonic

SMART WORKWEAR SYSTEM

User Manual

Ergonomic risk assessment & work
technique training, powered by wearable
motion sensors.

MANUAL

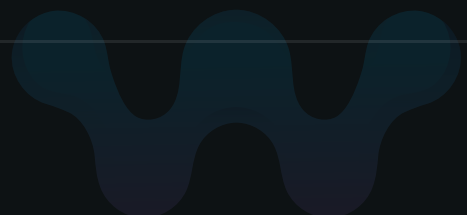
Version 3.1

MOBILE APP

Android v5.1.7

COVERS

App & Web Dashboard



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INTRODUCTION

The Wergonic System

Wergonic is a smart-workwear system for fast, accurate measurement of upper-body posture and motion during real work tasks — with results compared directly against recommended reference values.

Smart Workwear

- Motion sensors fitted into garments for easy on/off use
- Tracks head, trunk, both arms, hand and forearm
- Estimates reachability (hand's reach)

Mobile App · Android

- Streams and records sensor data over Bluetooth
- Real-time posture & movement metrics
- Annotate activities and breaks
- Syncs measurements to the web dashboard

Web Dashboard

- Programme work cycles, activities and conditions
- Generate risk-assessment reports
- Built-in frameworks: RAMP, RULA, REBA, MEC (Volvo)
- AI-assisted interpretation of results
- User & organisation admin tools
- Runs in any browser — **cloud.wergonic.com**

Your account. You receive credentials to sign in to both the app and the dashboard with an email of your choice. Within an organisation, colleagues' accounts are linked for shared work.

2 HARDWARE What's in the Kit

The Wergonic Smart Workwear is delivered as a standardised kit.

- **Six IMUs** in colour-coded moulds
- **Android phone** with the Wergonic App installed and pre-paired
- **Wergonic shirts** in six sizes (XS–XXL)
- **Wergonic glove** and **headband**
- **Six CR2025** coin-cell batteries + one **phone charger**
- **Heart-rate chest band** with a mould-free IMU (optional)

Phone compatibility

Use an Android phone on **Android 13 or newer**. Bluetooth hardware quality affects streaming. Tested models:

- Samsung Galaxy S21 FE · S21 Ultra
- Google Pixel 6A · Pixel 7A

Compliance: CE, FCC, IC · Bluetooth 4.0/5.0 · RoHS, REACH.



The six Wergonic / Movesense IMUs in their colour-coded moulds — Right arm Blue, Trunk Green, Head Black, Left arm Red, Hand Silver, Forearm Beige.

Sensors & What It Measures

The system uses six Movesense inertial measurement units (IMUs), held in colour-coded moulds for correct placement.

Sensor & specifications

- 9-axis motion (accelerometer, gyroscope, magnetometer) + temperature
- Accelerometer $\pm 2/4/8/16$ g · Gyroscope ± 125 – 2000 °/s
- Sampling 12.5–833 Hz · Bluetooth 4.0/5.0
- 36.6 mm $\varnothing$ × 10.6 mm · 9.4 g · CR2025 battery
- Waterproof & shockproof

Colour code

- Head — Black
- Trunk — Green
- Right arm — Blue
- Left arm — Red
- Hand — Silver
- Forearm — Beige

3.1 · Where each sensor goes

BODY PART	COLOUR	PLACEMENT
Head	Black	Centre of the forehead , held by the headband.
Trunk	Green	Upper back, at the base of the neck (cervical/thoracic junction), in the shirt's back pocket.
Right arm	Blue	Outer side of the right upper arm, over the deltoid, in the shirt's sleeve pocket.
Left arm	Red	Outer side of the left upper arm, over the deltoid, in the shirt's sleeve pocket.
Hand	Silver	Back of the dominant hand, in the glove's hand pocket.
Forearm	Beige	Dominant forearm, in the glove's ribbed cuff — see the photo in 5.2.

Forearm and wrist are not the same thing. There is no wrist sensor. The beige sensor is the **forearm** sensor — it sits on the forearm, close to the wrist, and that is what it is called everywhere in the app. The **wrist** is a result, not a sensor: it is calculated from the *Hand* and *Forearm* sensors together, as the difference in movement between them. That is why the report has a *Wrist* page even though no sensor is called Wrist.

3.2 · What is measured on each body part

All angles are inclinations measured against a **vertical reference** set during calibration. All speeds are **angular velocity** — how fast the body part moves, in degrees per second.

BODY PART	ANGLES (°)	SPEED OF MOVEMENT (°/S)
Head	Forward/backward bending · Side bending	Not reported

Trunk	Forward bending · Side bending	Not reported
Right arm Left arm	Elevation of the upper arm (all shoulder movements, relative to vertical) · Forward bending · Side bending	Arm angular velocity
Forearm	Elevation of the forearm — used together with the upper-arm angle to estimate how far the hand reaches from the body	Feeds the wrist result below
Hand	—	Vibration detection — a live indicator (%) of how sure the app is that the hand is exposed to vibration. Live only; not included in the report
Wrist (calculated)	Not reported as an angle	Wrist angular velocity — how fast the hand moves relative to the forearm, from the Hand and Forearm sensors together
Heart rate (optional band)	—	Heart rate, and time spent above 33% of heart-rate reserve

Rotation is not measured. The system does not measure axial rotation — twisting of the trunk, turning of the head, or rotation of the upper arm. Every angle in the report is an *inclination* against vertical. The only rotational figure is the wrist's angular velocity, which is the speed of the hand turning relative to the forearm, not a twist angle.

Live tiles vs report. During recording, head, trunk and arm *side bending* and arm *forward bending* are shown as extra tiles — switch them on with **Show extra tiles** in Settings. The full report is covered in chapter 6, and the recommended limits for each quantity in chapter 7.

4

GETTING STARTED

System Setup

Use the provided phone (mostly pre-configured) or your own Android phone. Either way, finish by setting up your Work Library on the dashboard.

4.1 · Using the provided phone

The kit phone already has the correct Android version, the latest Wergonic App and pre-paired IMUs. Two steps remain:

1. **Sign in to Google Play** with a Gmail account so the app stays updated with new releases.
2. **Set up your Work Library** on the dashboard (see 4.4).

Resetting the phone to factory settings is optional. If you do, follow the "own phone" steps below afterwards.

4.2 · Using your own phone

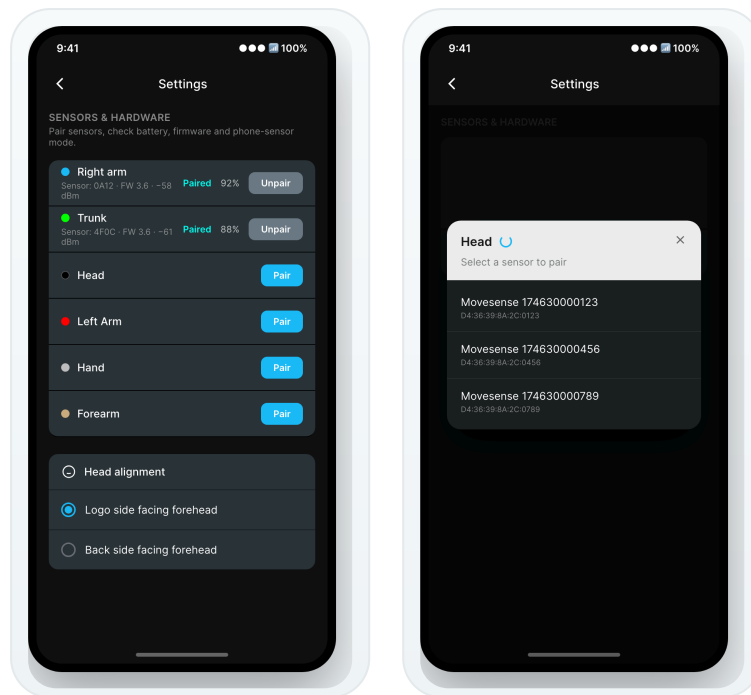
1. **Install the app** — open Google Play, sign in, search **Wergonic**, install the official app.
2. **Start the IMUs** — insert a CR2025 into each unit (pry the lid with a coin). A **red light blinks** when the IMU is on; it can take up to 30 seconds to appear.
3. **Sign in** to the app with your credentials.



Opening the battery lid and inserting a CR2025 coin cell — use a coin (or small spoon) to pry the lid, then reattach it.

4.3 · Pairing the IMUs

Open **Settings** (gear icon, top-right of the home screen) and go to **Sensors & Hardware**. Each body part has a **Pair** button; tap it and pick the sensor from the list. Identify a sensor by the last four digits of the serial number on its back. Once connected, the row shows **Paired** in cyan with battery level and an **Unpair** option.



A Settings → Sensors & Hardware. Pair / Unpair each body part; "Head alignment" sits below.

B "Select a sensor to pair" — tap the matching serial number to connect it to that body part.

Colour-code standard — Head Black · Trunk Green · Right arm Blue · Left arm Red · Hand Silver · Forearm Beige. The optional HR chest band pairs with the mould-free sensor.

Head alignment. The head IMU can face either way on the headband. In Settings, set **Logo side facing forehead** (logo turned inwards, towards the skin) or **Back side facing forehead** (logo turned outwards) to match how you fitted it.

Test auto-pairing. Log out, close and reopen the app — sensors should reconnect automatically. If you're not measuring right away, remove the batteries to save charge.

Phone settings (own phone)

- **Battery:** turn off adaptive/optimised modes; set the Wergonic App to **Unrestricted**.
- **Permissions:** Location and Nearby devices → **Allow all the time**.
- **Notifications:** enable for the Wergonic App.
- **Unused-apps / app-pausing:** disable for the Wergonic App.

4.4 · Set up your Work Library on the dashboard

Sign in at cloud.wergonic.com. In the left menu, under **Setup**, open **Work Library**. This is where you prepare everything the app offers you during a measurement. The page has three tabs:

TAB	WHAT IT HOLDS
Work Cycles	Your factories and the work they contain, as a tree: Factory → Line → Workstation → Operator → Work cycle → Tasks . Each task has a name, an optional description and a duration in seconds. Required when you measure a work cycle.

Activities	Activity groups and the activities inside them. These are the buttons you tap during a free recording to mark what the worker is doing. Optional, and only available when the phone is online.
Conditions	Condition groups and their values — the circumstances you want to tag a measurement with (single-choice or multiple-choice). Optional.

Adding work cycles. On the *Work Cycles* tab use **Add factory** to build the tree by hand, then add lines, workstations, operators, work cycles and tasks step by step. To load many at once, prepare them in Excel and use **Import CSV/Excel**.

Adding activities and conditions. On the *Activities* or *Conditions* tab, first create a group, then add items to it. Items can later be renamed, merged into another item, or retired.

Menu names. The left menu is grouped into **Overview** (Home), **Measurements**, **Risk assessments**, **Workers**, **Setup** (Work Library) and — for organisation admins — **Team**. This manual uses those exact names.

Running a Measurement

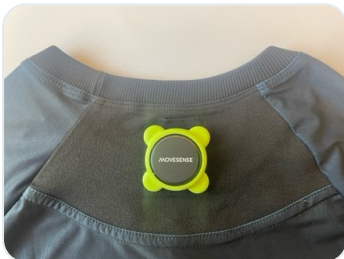
5.1 • Prepare

- Fully charged phone and six paired IMUs
- Fresh CR2025 batteries (recommended)
- Clean Wergonic shirt & glove in the right size, plus a headband for the head IMU
- A ~2 kg weight for arm calibration

5.2 • Start the IMUs & fit the workwear

Insert batteries and confirm each IMU shows a **red blinking light**. Unless stated otherwise, fit each sensor with the Movesense logo **facing outward and upright**:

- **Head (black)** — **headband**. Put the headband on so the sensor sits in the **centre of the forehead**, just above the eyebrows and below the hairline. *Not* over the ear and not on the back of the head. The sensor may face either way; whichever you choose, set **Head alignment** in Settings to match (see 4.3).
- **Trunk (green)** — **shirt**. Slide the sensor into the pocket on the **upper back**, at the base of the neck. The shirt should fit snugly so the sensor cannot slide.
- **Arms (blue, red)** — **shirt sleeves**. Right arm blue, left arm red, in the pocket on the outer side of each upper arm.
- **Hand (silver)** and **Forearm (beige)** — **glove**. Silver on the back of the hand, beige in the ribbed cuff that sits on the forearm. Both are worn on the **dominant** arm.



Trunk IMU seated in the shirt's upper-back pocket at the base of the neck — logo facing outward and upright.

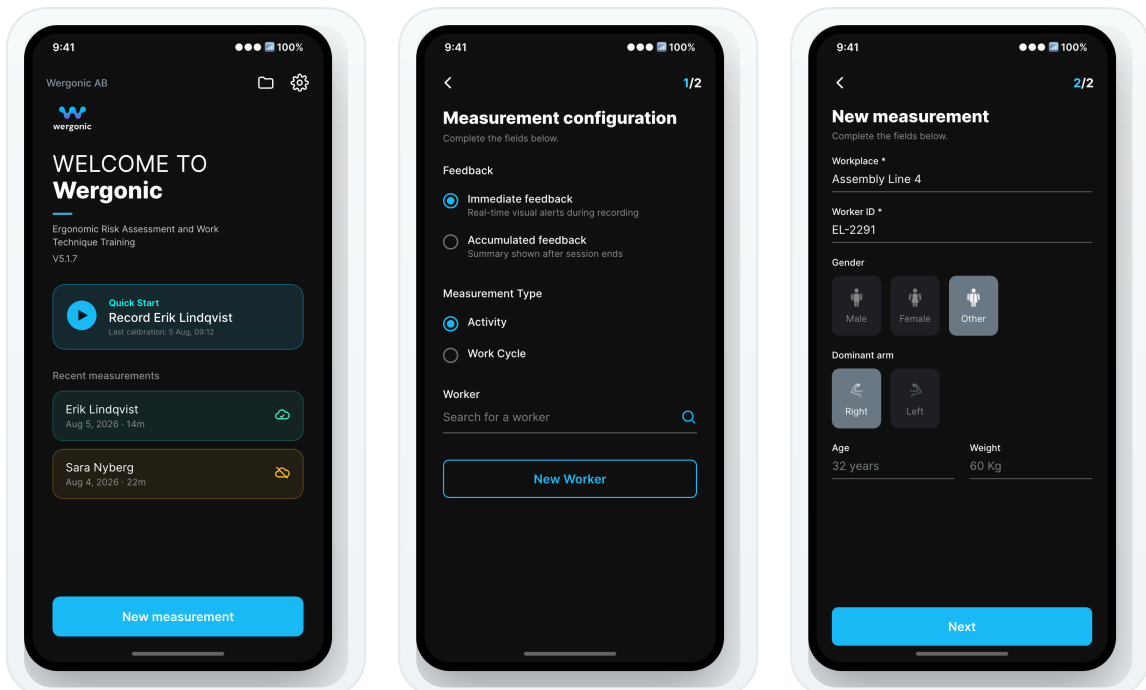


The glove, laid flat. Beige **Forearm** sensor in the ribbed cuff (left), silver **Hand** sensor on the back of the hand (right).

The hand and forearm sensors work as a pair. Connect both, or neither. The wrist result is calculated from the two together — with only one of them connected there is no wrist result at all.

5.3 · Start a new measurement in the app

On the home screen tap **New measurement**, then complete the two setup pages and confirm pairing before calibration.



C Home — tap **New measurement**. The folder icon opens history; the gear opens settings.

D Page 1 — two separate choices: **Feedback** (what the coloured squares react to while recording) and **Measurement Type** (how the work is logged). Then pick or search a worker.

E Page 2 — worker profile (workplace, ID, gender, dominant arm, age, weight). Tap **Next**.

Page 1 in detail — Feedback and Measurement Type are independent

They answer two different questions. You set both, in any combination.

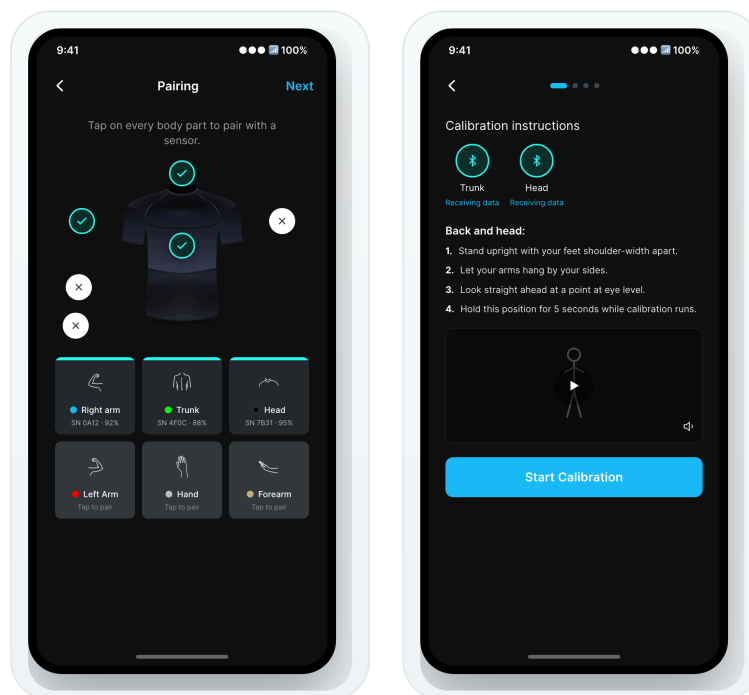
SETTING	OPTION	WHAT IT DOES
Feedback What the coloured squares react to while recording	Immediate feedback	The square reacts to the posture right now . Go past a limit and it turns orange or red straight away, and vibration or voice cues fire if you switched them on in Settings. Use this for work-technique training, where the worker should feel the effect of each movement.
	Accumulated feedback	The square reacts to how much of the time so far has been spent in a risky posture, not to the current movement. It stays grey for the first minutes while it gathers data — 10 minutes by default, adjustable in Settings — and shows a countdown. After that it turns orange or red if too large a share of the measured time has been spent past a limit. Use this to judge the job as a whole rather than a single movement.

Measurement Type	Activity	
How the work is logged		Record freely and log activities as they happen. Nothing is planned in advance — you tap the activity being done now, and the previous one ends automatically.
	Work Cycle	Follow a predefined task list for a station. Pick factory, line, workstation, operator and model first, then step through the tasks manually or let the app advance them automatically. Requires a work cycle prepared in the Work Library, and an online phone.

Online vs offline. To load a pre-programmed **Work Cycle** or reuse a saved worker profile you must be online. Offline, create a new profile and sync later.

5.4 · Confirm pairing, then calibrate

The **Pairing** screen shows the shirt with each body part. Paired sensors show a **cyan check**; tap a body part to pair, or tap a paired one to unpair. When ready, tap **Next** (top-right) to calibrate.



F Pairing overview — connection cards show each sensor's serial and battery. Tap **Next** when the sensors you need are connected.

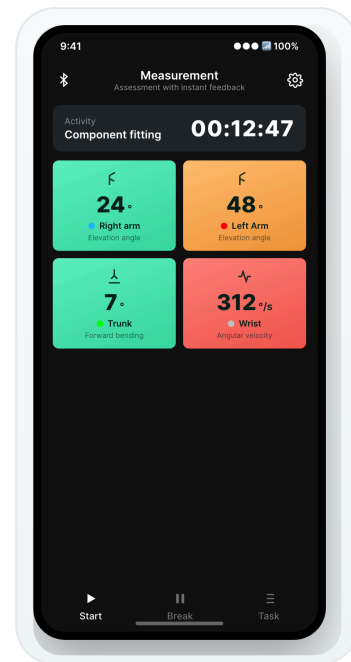
G Calibration — follow the numbered steps and demo video, then **Start Calibration** and hold the pose for 5 seconds.

Calibration is done per test person, in **four steps** — **Back and head**, **Right arm**, **Left arm**, and a **Forward reference**. Each step gives text, a demo video (with audio) and voice cues; the arm steps use the ~2 kg weight. Calibration is saved on the phone and can be reused for consecutive measurements on the same person **if the IMUs have not moved** — otherwise recalibrate.

5.5 · Live mode & recording

After calibration you enter live mode. Metric squares update in real time and colour by risk — **green** (safe), **orange**, **red**.

- **Choose metrics:** long-press any square, tick the metrics you want, then **Save** (top-left).
- **Recalibrate:** before starting (timer at 00:00:00), tap the gear → a body part.
- **Record:** tap **Start**; use **Break** to pause; tap **Finish** to end and save.
- When online you can also add activities during recording.



H Live mode — Start / Break / Finish controls, with risk-coloured metric squares.

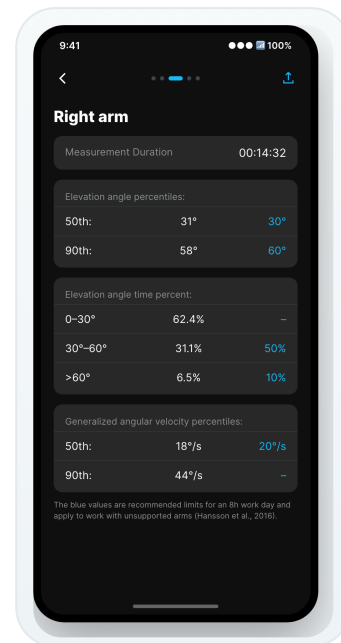
Reviewing Results

Review a measurement directly in the app, or in full on the web dashboard.

In the app

Tap the **folder icon** (home, top-right) to open your measurements. On a measurement, tap the **: menu** → **Report**. The report opens as **swipeable pages** — one per body part — with the numbers in tables. Recommended limits are shown in **blue** next to your result; all of them are listed in chapter 7.

- The **: menu** also offers **Sync measurement**, **Share** and **Delete**.
- Tapping a measurement opens its **timeline** (activities, breaks).



i In-app report — per-body-part tables; blue values are the recommended 8-hour limits.

On the web dashboard

Online measurements upload automatically; offline ones upload when you reconnect (may take a few minutes). Sign in at **cloud.wergonic.com** and open **Measurements** in the left menu. The table lists every measurement with its worker, recorded body parts, duration, start time, risk assessments and status. **Click a finished measurement** to open its details.

Results are shown as:

- **Pie charts** — posture (°) first; switch to *Speed of movement* for angular velocity (°/s). Export to Excel/PDF, or open *Show Detailed View* for time plots.
- **Tables** — posture, an algorithm-based reach estimate, then speed of movement.

The row menu on each measurement also offers **Run risk assessment**, **Generate report**, **Export raw data** and **Archive**. Select several measurements to **compare** them or build a **combined report**.

7 INTERPRETATION Reference Values

These are the recommended limits the app shows in blue beside your results. They apply to a full **8-hour working day**. Exceeding one is not a diagnosis — it is a signal that the exposure should be reduced.

7.1 · How to read the numbers

- **50th percentile (median)** — the value the worker was at or below for half of the measured time. This is the typical working posture or speed.
- **90th percentile** — the value reached during the most demanding 10% of the time. This captures the peaks.
- **Time percent** — how much of the measured time was spent inside each angle band.

7.2 · Recommended limits

BODY PART	QUANTITY	RECOMMENDED LIMIT	SOURCE
Arms (unsupported)	Elevation angle — 50th percentile	≤ 30°	Hansson et al., 2016
	Elevation angle — 90th percentile	≤ 60°	
	Angular velocity — 50th percentile	≤ 30 °/s	
Trunk	Forward bending — 50th percentile	≤ 20°	Hansson et al., 2016
	Side bending — 50th percentile	≤ 20°	
Head	Side bending — 50th percentile	≤ 20°	No source shown in the app
Wrist	Angular velocity — 50th percentile	≤ 20 °/s ≤ 15 °/s if the work is also assessed as force-demanding	Arvidsson et al., 2017
Heart rate (optional band)	Accumulated time above 33% of heart-rate reserve (HRR)	Reported as a figure; the app shows no recommended limit	—

The limits above are the ones the app displays. Head forward bending is reported without a limit value.

7.3 · Angle bands used in the report

The report also shows how much of the time was spent in each band. Anything below 10° counts as neutral and is not listed. Leaning backwards more than 5° is counted separately — the accumulated feedback uses it during recording, but it is not shown in these tables.

QUANTITY	BANDS REPORTED
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Trunk — forward bending	10°–20° · 20°–45° · > 45°
Trunk — side bending	0°–10° · 10°–30° · > 30°
Head — forward bending	10°–20° · 20°–30° · > 30°
Arm — elevation	0°–30° · 30°–60° · > 60°

7.4 · Full references

The app cites its sources in short form. Written out in full, they are:

1. Hansson, G.-Å., Arvidsson, I., & Nordander, C. (2016). *Riktvärden för att bedöma risken för belastningsskador, baserade på tekniska mätningar av exponeringen* [Guideline values for assessing the risk of work-related musculoskeletal disorders, based on technical measurements of the exposure]. Arbets- och miljömedicin, Lund, Report No. 4/2016.
Cited on the arm and trunk pages of the report. The limits apply to work with unsupported arms.
2. Arvidsson, I., Dahlgvist, C., Enquist, H., & Nordander, C. (2017). *Åtgärdsnivåer mot belastningsskada* [Action levels against work-related musculoskeletal disorders]. Arbets- och miljömedicin Syd, Report No. 18/2017. Lund, Sweden.
Cited on the wrist page of the report.



CARE

Maintenance & Return

Between sessions

- **Remove the IMUs**, then machine-wash garments at **30 °C** with regular, perfume-free detergent (no disinfecting softener).
- **Disinfect** the removed IMUs.
- Avoid fully discharging the phone.

Returning the kit

- Clean the workwear as above.
- Batteries needn't be returned — recycle any you won't reuse.
- Fully charge the phone.
- Log out of Google Play and the Wergonic App.



Troubleshooting & FAQ

Troubleshooting

PROBLEM	WHAT TO DO
IMU won't blink / seems off	Moist fingers can discharge coin cells. Try a fresh battery; if it still won't start, contact Wergonic for a replacement.
IMU is slow to pair	Start the IMU <i>before</i> opening the app and wait ~1 minute. Don't repeatedly tap pair/unpair. Try another location — some environments disturb Bluetooth.
Battery lid won't open	Use only CR2025, inserted the right way up. Never use CR2032 — it can jam and damage the lid.
App stuck loading after calibration	Usually an offline network search. Enable flight mode <i>with Bluetooth on</i> , or work online for best stability.
No wrist result in the report	The Hand and Forearm sensors work as a pair. If only one of them was connected, no wrist value can be calculated.

Frequently asked questions

Do I need to be online to measure?

No — only Bluetooth is required. Data is stored on the phone and uploads when you're back online. Online mode is more stable and is recommended.

How far does Bluetooth reach?

Roughly 10 metres, but range depends on the environment where you measure.

Does the system measure rotation?

No. It measures inclination angles against vertical and speed of movement. Twisting of the trunk, head or upper arm is not measured — see chapter 3.2.

Why does the app say "Forearm" but the report says "Wrist"?

Because they are two different things. The beige sensor sits on the forearm and is called Forearm everywhere it is set up. There is no wrist sensor — the wrist figure is calculated from the Hand and Forearm sensors together. See chapter 3.1.

Where exactly does the head sensor go?

In the centre of the forehead, held by the headband — not over the ear. See chapter 5.2.

Need help? Contact the Wergonic team for hardware replacements or support with setup and measurements.